

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
CO4_AT3 — TIME-SERIES & GEOSPATIAL ANALYSIS REPORT

Course Code:	DSA06	Course Title:	Data Handling and Visualization
CO Assessed:	CO4 – Evaluation (BL5)	Assessment:	Assessment Tool 3 – Time-Series & Geospatial Analysis Report
Weightage:	20% of CIA	Total Marks:	20 (2 Questions × 10 Marks)
CO4 Statement:	<i>Evaluate real-world time-series and geospatial visualizations, analyzing trend/seasonality/noise and design effectiveness, and justify recommendations.(BL5)</i>		
Student Name:	Kola Amrutha Sandhya	Reg. No.:	192524270
Section / Batch:	AIDS	Date:	22-08-2026

Code of Conduct

I Kola Amrutha Sandhya 192524270 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Amrutha

Instructions

- This test contains 2 questions — one Stock Analysis problem and one Geospatial Analysis problem — each carrying 10 marks. Total: 20 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 90 minutes.
- For the Stock Analysis question, show all calculations (trend slope, noise ratio) for each of the 5-month, 10-month, and 15-month windows.
- For the Geospatial Analysis question, your evaluation must explain WHY the design choice succeeds or fails, and your recommendation must be specific — not just “use a different chart.”
- Diagrams may be hand-drawn or clearly described in words if drawing tools are unavailable.

Annexure A – TIME-SERIES & GEOSPATIAL ANALYSIS REPORT

Rubrics for Calculation **ASSESSMENT RUBRIC**
AT3 — Time-Series & Geospatial Analysis Report

CO4 · BT5: Evaluate ·

This rubric is used to assess student responses to the CO4-AT3 Time-Series & Geospatial Analysis Report, in which students analyze a stock's trend/seasonality/noise across multiple time windows, and evaluate a geospatial visualization's design choices, justifying their conclusions.

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates thorough understanding of trend/seasonality/noise analysis across time windows, and of the relevant geospatial visualization principle being evaluated.	Good understanding of both areas, with only minor gaps.	Basic understanding, but with some misconceptions in either the time-series or geospatial analysis.	Poor understanding of core concepts in one or both areas.
Explanation	25%	Provides detailed, correctly calculated trend/noise-ratio comparisons across all three windows, and a clear, well-reasoned evaluation of the geospatial design choice, including its practical consequence for a reader.	Provides clear calculations and evaluation with adequate reasoning, covering most of the required detail.	Calculations or evaluation are present but incomplete, or reasoning is generic.	Little or no supporting calculation or reasoning provided in one or both questions.
Application	20%	Effectively applies analysis techniques to reach a well-supported conclusion on window reliability, and proposes a specific, well-justified	Applies techniques to reach a reasonable conclusion and recommendation, with only minor	Limited application; conclusions or recommendations are only partially supported.	Fails to apply appropriate techniques; conclusions and recommendations are unsupported or

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
		geospatial recommendation.	errors.		incorrect.
Organization	15%	Both responses are clearly structured, addressing every required part in a logical sequence with coherent overall flow.	Both responses are organized, with only minor issues in flow or clarity.	Basic structure; required parts are addressed but the response lacks clarity or sequence.	Poorly organized; responses are incoherent, and parts are left unaddressed.
Accuracy	10%	All parts of both answers — calculations, evaluation, and recommendation — are completely accurate and well-suited to the specific data and scenario given.	Mostly accurate, with only minor omissions or a small error in one question.	Partially correct; one or both answers contain notable inaccuracies.	Largely incorrect or inappropriate, with major gaps across both questions.

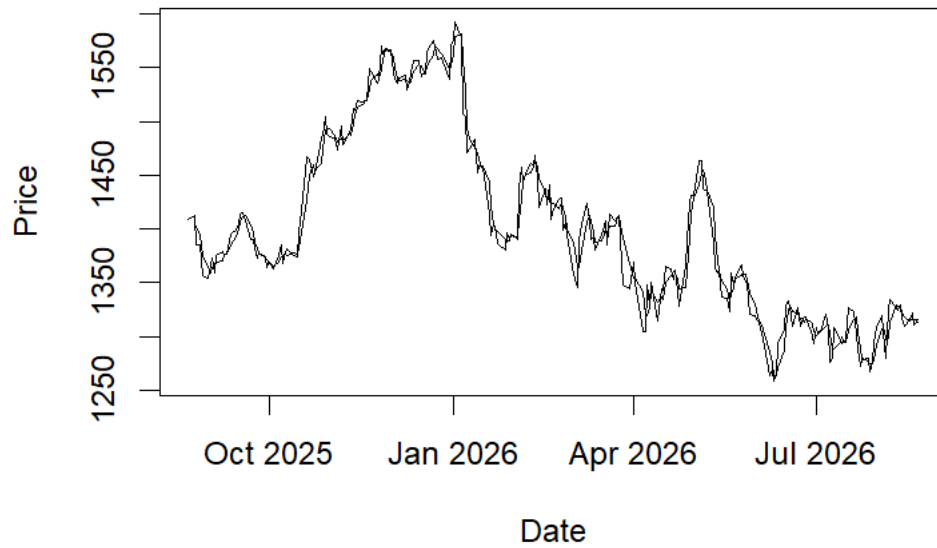
Scoring Guide

For each criterion, award a performance level (1–4) based on the descriptors above. Multiply the level achieved (as a percentage of 4, i.e., Level 4 = 100%, Level 3 = 75%, Level 2 = 50%, Level 1 = 25%) by the criterion's weightage, then sum across all five criteria to obtain the total score out of 100%.

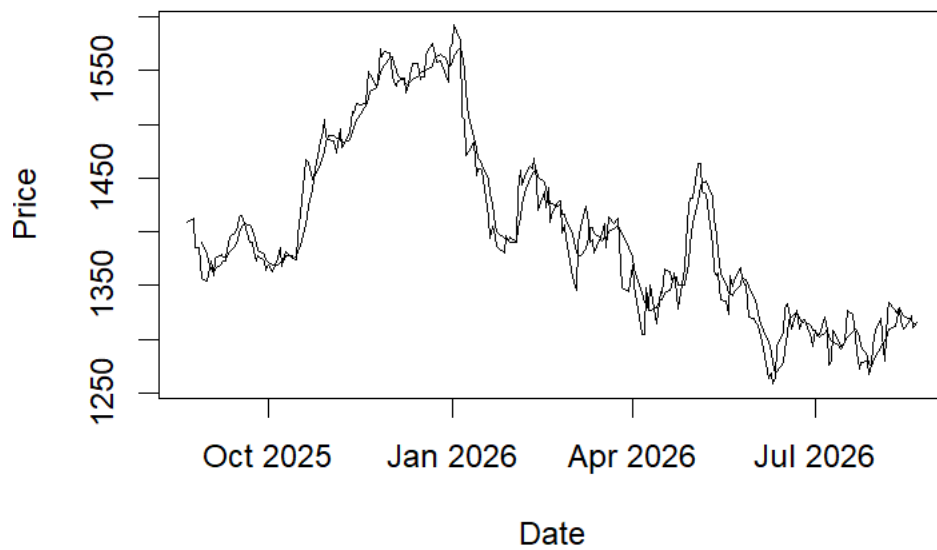
Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

1. RELIANCE STOCK

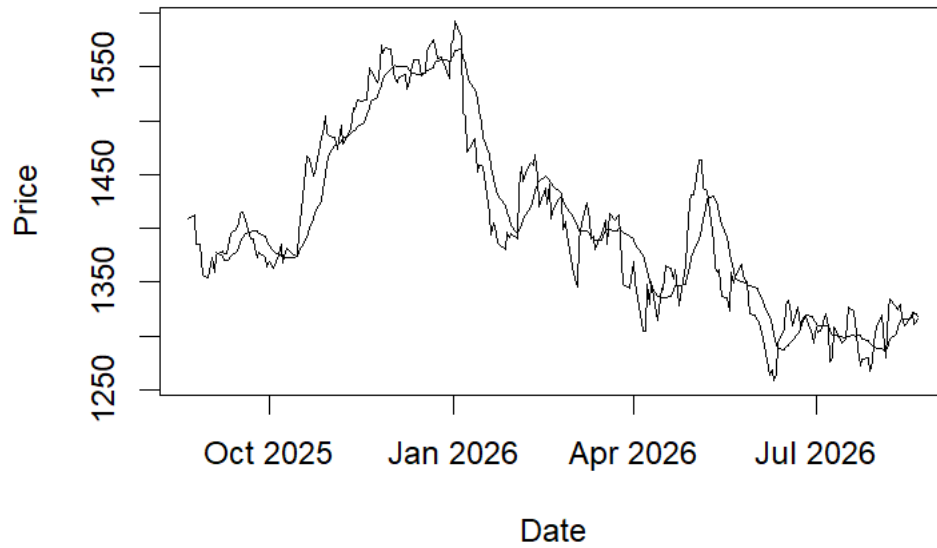
3-Day Moving Average



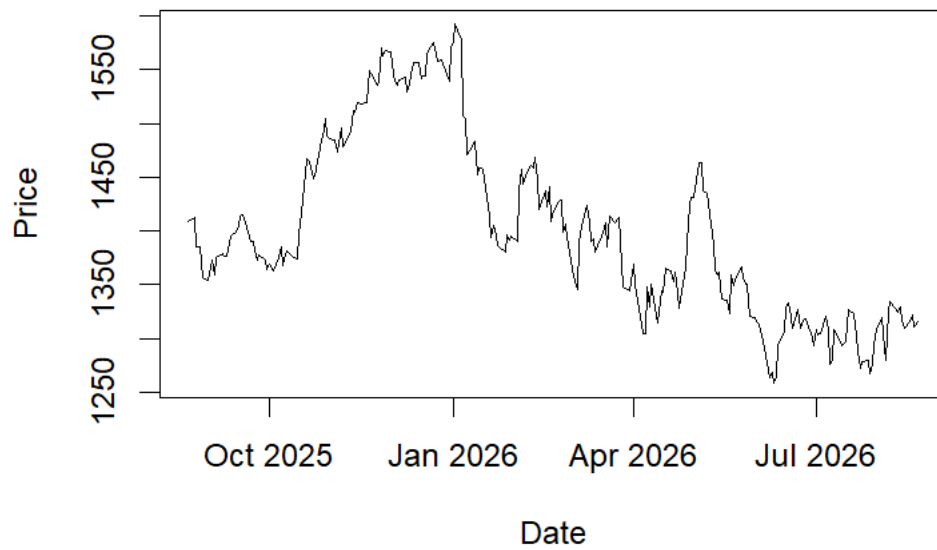
5-Day Moving Average



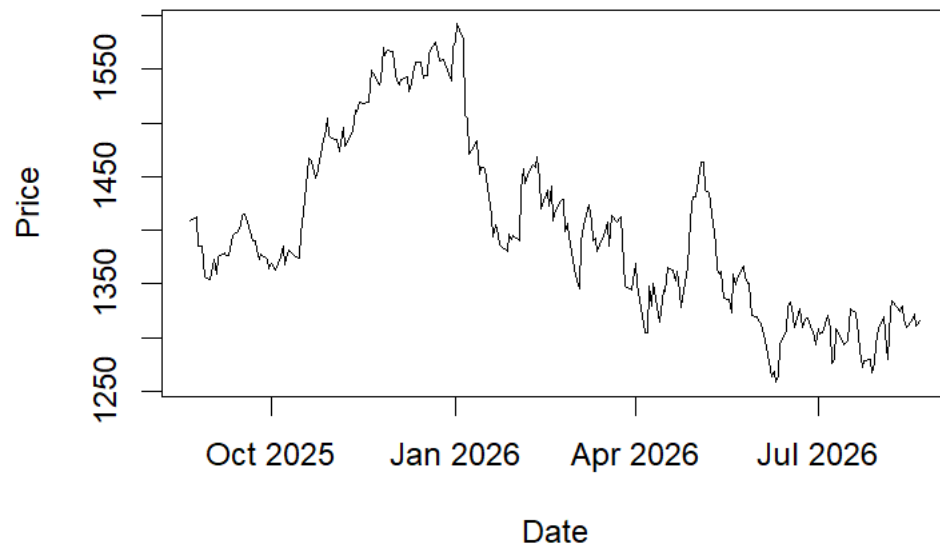
2-Week Moving Average



1-Month Moving Average



6-Month Moving Average



Reliance Industries - 1 Year



2. GEOSPATIAL

3.

